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21 JANUARY 1963
SUPERSEDING
MIL–P–46572(ORD)
25 AUGUST 1961

MILITARY SPECIFICATION

PRIMER, PERCUSSION-ELECTRIC, M67 PARTS FOR

1. SCOPE

1.1 This specification covers the component parts for one type of primer designated as Primer, Percussion-Electric, M67, for use in artillery ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 —Ammunition and Special Weapons; General Specification for

MIL–P–10025 —Packaging, Packing and Marking for Interplant Shipment

MIL–I–45208 —Inspection Requirements; General Specification for

MIL–I–45607 —Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes

MIL–STD–109 —Quality Assurance Terms and Definitions

DRAWINGS

ORDNANCE CORPS

7548469 —Primer, Metal Parts Assembly

PUBLICATIONS

ORDNANCE CORPS

ORD–M608–11 —Procedures and Tables for Continuous Sampling by Attributes

IEL–7548469 —Index of Inspection Equipment Lists

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials shall be in accordance with applicable drawings and specifications.

* 3.2 **Metal parts assembly.** The meal parts assembly shall comply with all requirements specified on Drawing (dwg.) 7548469 and with all requirements specified in applicable specifications.

3.3 **Hydrostatic test.** The primer tubing shall withstand the application of a minimum (min.) hydrostatic pressure without rupture or indication of flaws, as evidenced by leaks, and without permanent deformation beyond drawing requirements, when tested as specified in 4.4.1 (see 6.5).

3.4 **Pull test of brazed joint.** The brazed joint securing the head to the primer body shall withstand a pull of 5500 pounds, min. without evidence of separation of the head from the primer body when tested as specified in 4.4.2.

3.5 **Workmanship.** All parts shall be dry and free of chips, dirt, grease, rust and other

* See 6.4.

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foreign material. Particular attention shall be given to freedom of the parts from burrs, sharp edges, and defects in metal such as cracks, porosity, pipes, etc. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

* **4.2 Initial production inspection.**

4.2.1 *Submission.* Prior to normal production the contractor shall submit an initial production sample consisting of 25 each of every individual component and every subassembly, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production The

* See 6.4.

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initial production sample shall be accompanied by certificates of analyses for all materials. A sample shall also be submitted whenever a change occurs in manufacturing process, material used, drawing specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 *Inspections to be performed.* Each subassembly and component shall be inspected by the Government for all the inspections specified in 4.3 and 4.4 of this specification and for all additional requirements of the applicable drawings.

4.2.3 *Rejection.* If any subassembly or component fails to comply with the applicable requirements, the initial production sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the contracting officer.

* **4.3 Inspection provisions.**

4.3.1 *Lot formation.* A lot shall consist of metal parts assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number.

4.3.2 *Examination.* Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL–STD–105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD–M608–11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Head (see dwg. 7548316 covering a detail of dwg. 7548469).

Categories	*Defects*	*Method of inspection*	*Code No. (see 6.2)*
Critical: None defined			
Major:	AQL 0.25 percent		
101.	Crack or fracture	Visual	01001
Minor:	AQL 1.50 percent		
201.	Total length, min.	Gage	01002
202.	Depth of body counterbore, maximum (max.)	Gage	01003
203.	Any operation missing	Visual	01004
204.	Evidence of poor workmanship (see 3.5)	Visual	01005

4.3.2.2 Body (see dwg. 7548315 covering a detail of dwg. 7548369).

Categories	*Defects*	*Method of inspection*	*Code No.*
Critical:			
1.	Flash hole missing	Gage	02001
2.	Crack or split	Visual	02002
Major: None defined			
Minor:	AQL 2.50 percent		
201.	Location of first row of holes	Gage	02003
202.	Wall thickness	Gage	02004
203.	Diameter of flash holes	Gage	02005
204.	Total length	Gage	02006
205.	Outside diameter, max.	Gage	02007
206.	Any operation missing	Visual	02008
207.	Evidence of poor workmanship (see 3.5)	Visual	02009

* See 6.4.

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4.3.2.3 Assembly (see dwg. 7548469).

Categories	Defects	Method of inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Pitch diameter of external thread, min.	Gage	03001
102.	Pitch diameter of internal thread, max.	Gage	03002
103.	Major diameter of external thread, min.	Gage	03003
104.	Minor diameter of internal thread, max.	Gage	03004
105.	Braze not full 360 degrees	Visual	03005
106.	Solder on interior	Visual	03006
107.	Protective coating missing or incomplete	Visual	03007
Minor:	AQL 6.50 percent		
201.	Pitch diameter of external thread, max.	Gage	03008
202.	Pitch diameter of internal thread, min.	Gage	03009
203.	Thickness of flange	Gage	03010
204.	Diameter of flange, max.	Gage	03011
205.	Depth of interior thread counterbore	Gage	03012
206.	Diameter under flange, max.	Gage	03013
207.	Diameter below external thread	Gage	03014
208.	Inside diameter, min.	Gage	03015
209.	Concentricity of body counterbore with thread	Gage	03016
210.	Concentricity of internal thread with external thread	Gage	03017
211.	Concentricity of flange with external thread	Gage	03018
212.	Concentricity of counterbore with internal thread	Gage	03019
213.	Length from base to end of thread, min.	Gage	03020
214.	Length of external thread undercut, max.	Gage	03021
215.	Diameter of internal thread counterbore	Gage	03022
216.	Depth from counterbore to end of internal threads, min.	Gage	03023
217.	Width of wrench slot, min.	Gage	03024
218.	Total length, max.	Gage	03025
219.	Marking misleading or unidentifiable	Visual	03026
220.	Any component or operations missing	Visual	03027
221.	Evidence of poor workmanship (see 3.5)	Visual	03028

* 4.3.3 *Testing.*

4.3.3.1 *Hydrostatic test (see 3.3)*—Critical defect—Code No. 02011. This test shall be performed 100 percent as specified in 4.4.1. If any primer tubing fails to comply with the applicable requirement, it shall be classed defective and no primer bodies shall be fabricated therefrom.

4.3.3.2 Pull test of brazed joint (see 3.4). Critical defect—Code No. 03029.

4.3.3.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, the acceptance and rejection criteria for this test shall be in accordance with Standard MIL–STD–105, table IV A, code letter L and an AQL of 0.065 percent. The test shall be performed as specified in 4.4.2.

4.3.3.2.2 After the three consecutive lots have met the criteria of 4.3.3.2.1, the acceptance and rejection criteria shall be in accordance with Standard MIL–STD–105, table IV A, code letter I and an AQL of 0.25 percent.

* 4.3.4 *Inspection Equipment.* Index of Inspection Equipment Lists Number IEL–754-8469 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production

* See 6.4.

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and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.4 Test methods and procedures.

4.4.1 *Hydrostatic test.* Prior to machining, the primer tubing shall be tested for not less than five seconds at the minimum hydrostatic pressure as calculated by the following formula:

P = 2 times S times t times .9 divided by D.

where:

P = test pressure in pounds per square inch (psi.).
S = minimum specified yield strength, psi.
t = specified minimum wall thickness in inches.
D = specified maximum outside diameter in inches.

Observation shall be made for rupture, or indication of flaws as evidenced by leaks, and for permanent deformation beyond drawing requirements.

4.4.2 *Pull test of brazed joint.* The assembly shall be assembled to the applicable test fixture specified in 4.3.4. The specified load shall be applied in an axial direction. The rate of travel of the testing machine shall be 0.100 to 1.125 inch per minute. Any assembly which fails to comply with the specified requirement shall be classed defective.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level C.* Packing and marking shall be in accordance with Specification MIL–P–10025.

* See 6.4.

* **5.2 Data cards.** Data card information shall be prepared for each lot in accordance with information specified in Specification MIL–A–2550.

* 6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the number of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design." Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Ordnance Designs.* Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design

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when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.3 *Specification control drawings.* Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Commercial items.* Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 *IEL codes.* The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF —Contractor's design responsibility on Ordnance format; ORDM–608–12.
CDCF —Contractor design responsibility on contractor format.
OD —Ordnance design.
ODMU —Ordnance design, mandatory for use.
CE —Commercial equipment.
SCD —Specification control drawing.

6.3.6 *Use of standard measuring equipment.* If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 *Submission of contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

* **6.4 Changes from previous issue.** The outside margins of this specification have been marked with asterisks to indicate where changes (deletions, additions, etc.) from the previous issue have been made. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship of the last previous issue.

6.5 Hydrostatic test. If the primer tubing procured under this specification has been previously subjected to the hydrostatic test specified in this specification, a second hydrostatic test will not be required.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation

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